CHAPTER 15

Optical Properties of Structures of Reduced Dimensionality

15.1 Quantum Wells – Quantum Size Effects

- The layer with lz = 400nm is bulk-like and shows the 1s exciton absorption peak followed by its ionization continuum (Transitions between valence and conduction band)
- With the width of QW L_z decrease, the absorption edge shifts to higher energy
- The absorptions of the various subband transitions appear (labelled by n_z =1, 2, . . .) and the splitting into heavy hole and light hole excitons, caused by the different mass dependent quantization energies

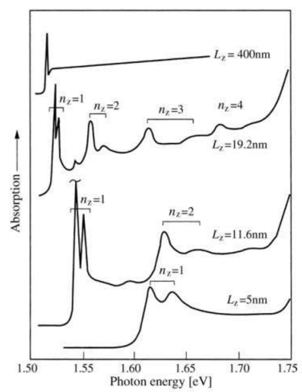


Fig. 15.1. Low temperature absorption spectra of GaAs (quantum) films of various thicknesses ([83G1,90G1] or [01L1] of Chap. 1)

- Fig. 15.2 We compare the spectra for GaAs/$Al_{1-y}Ga_yAs$ and $In_{1-y}Ga_yAs$/InP multiple quantum well structures.
- The prominent structures are n_z=1 hh and lh, n_z = 2 hh and lh. Higher states are usually less clearly seen.
- In contrast to the InP system, the GaAs substrate of the GaAs/AlGaAs MQW is opaque for the exciton resonances. So it has to be removed by selective etching for the observation of absorption spectra.

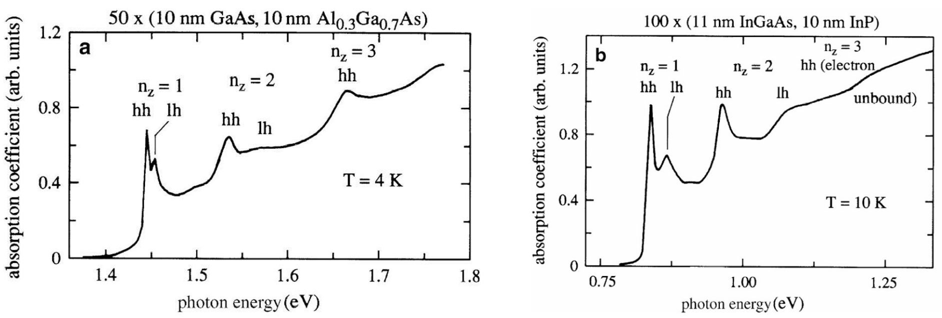


Fig. 15.2 The low temperature absorption spectra of multiple quantum well samples of GaAs/$Al_{1-y}Ga_yAs$ (a) and of $In_{1-y}Ga_yAs$/InP (b)

- The optical selection rules which arise when the excitons are confined to a QW:

$$\Delta n_z = 0$$

- The luminescence spectra of excitons in QW are often rather broad and partly Stokes-shifted with respect to the absorption.(Fig. 15.3)
- In samples of very high quality, resulting simultaneously in a reduction of the spectral widths of absorption and emission bands and of the Stokes shift.
- Due to the weaker coupling of carriers and excitons to optical phonons in the III–V compounds compared to the more ionic II–VI semiconductors, LO-phonon replica are hardly detectable in III-V material

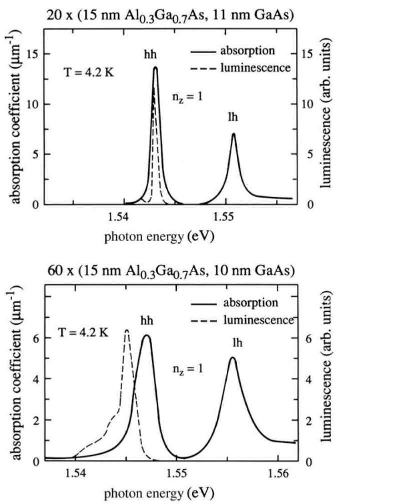


Fig. 15.3. A comparison of absorption and luminescence spectra in two samples of different quality [92O2]

Photoluminescence of BEC in QWs

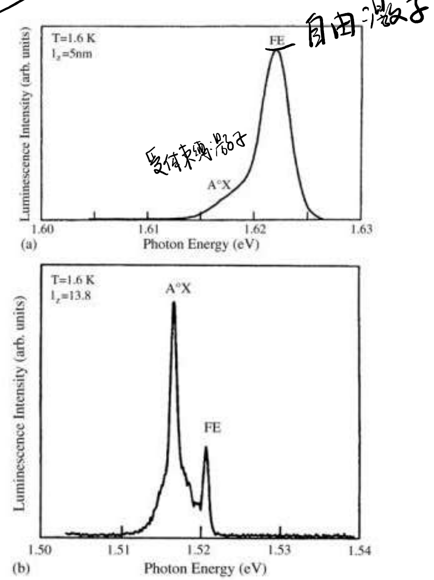


Fig. 14.7. The luminescence of $GaAs/Al_{1-y}Ga_yAs$ structures, of two different well width l_z center doped with Be acceptors at low temperature showing the emission from localized intrinsic excitons labelled FE and of acceptor bound excitons (A^0X) [89H1]

BEC also exists in quantum wells. Partly their luminescence merges with that of the so-called free, i.e., generally disorder broadened intrinsic emission lines as is often the case in $GaAs/Al_{1-y}Ga_yAs$ structures, partly it appears as a spectrally separated line.(Fig.14.7)

Assuming:

1. spherical dots
2. simple isotropic
3. parabolic conduction and valence bands
4. infinite barriers.

- Regime I:
  - Only the center of mass motion of the exciton is quantized, while the relative motion of electron and hole are hardly affected.
  - The quantization energy:

$$\Delta E_Q^{\mathrm{I}} = \frac{\hbar^2\pi^2}{2M\bar{R}^2} \qquad (15.7b)$$

- regime II:
  - The motion of the electron is quantized and the hole moves in the potential of the QD and of the space-charge of the electron for the usual situation $m_e < m_h$.
  - $$\Delta E_Q^{\mathrm{II}} = CRy^*\left(\frac{a_B\pi}{\bar{R}}\right)^2; with \quad C = 0.67 \qquad (15.7)$$

- Regime III:
  - $$\frac{\hbar^2\pi^2}{2\mu\bar{R}^2} = E_Q^{\mathrm{III}} >> E_{Coulomb}^{\mathrm{III}} \qquad (15.7f)$$
  - $$\Delta E(R) = \frac{\hbar^2\pi^2}{2\mu\bar{R}^2} - \frac{1.78e^2}{\varepsilon\bar{R}} + 0.752E_{Ry} \qquad (15.7g)$$

- Quantity affecting optical properties
  - The size of the dots
  - Annealing process of the glasses
  - The capping of dots

- We would expect QD to have an absorption spectrum consisting of a series of δ-functions or of Lorentians, if some homogeneous broadening is included.
  - Fig. 15.19 the size distribution of the Crystallites contribute to a broadening of the resonances.

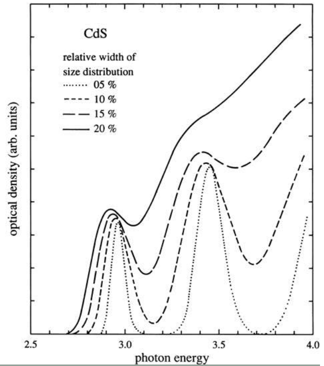
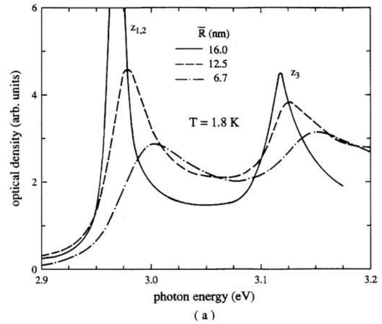
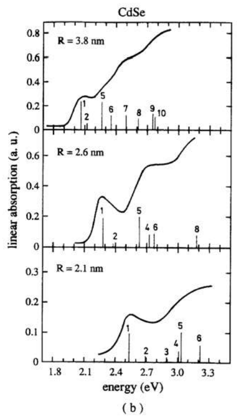


Fig. 15.19. Calculated spectra of the optical density of quantum dots for various relative widths of their size distribution.

- The range of the size distribution:
- For nanocrystallites in glass matrices:10 ~ 30%.
- For nanolithography:　(20 ± 10)%.
- Quantum islands in MBE-grown layers usually show strong size and composition fluctuations.

Fig. 15.20. A set of absorption spectra of CuBr quantum dots with different average radii $\bar{R}$ (a) *and a set of measured absorption spectra together with the calculated* positions of the optical transitions in CdSe QD (b)





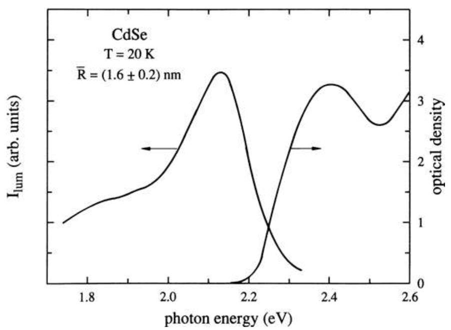


Fig. 15.21. The Stokes-shift between the absorbtion and emission spectra of $Cd_{1-x}Se_x$ quantum dots

- Fig. 15.23 To demonstrate the relation of inhomogeneous broadening of the spectrum of a dot ensemble and to the homogeneous broadening of a single dot.
  The satellite shifted by 25.6meV is the LO phonon replica.
- The absolute photo-luminescence quantum efficiency can be rather high, reaching values of 0.*1 to 0.9.*
- The homogeneous width *Γ(T):*

$$\Gamma(T) = \Gamma_0 + \frac{\Gamma_{el-phon}}{\exp\{\hbar\Omega_{LO}/k_B T\} - 1} \quad (15.9)$$

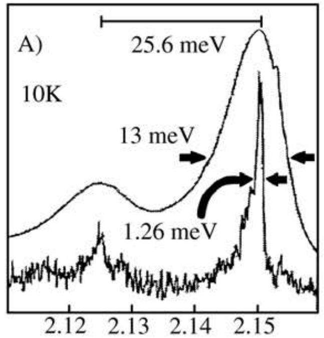


Fig. 15.23. Luminescence spectra of ZnS capped CdSe quantum dots for an ensemble of average radius R = 4.5 nm and of a single dot.